DEGRADX: A DATASET-CALIBRATED SYNTHETIC TESTBED FOR ATTRIBUTION EVALUATION ON DEGRADING SYSTEMS *

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ABSTRACT

Attribution methods are increasingly applied to models of degrading systems. On measured data, however, the contribution of each observation to a model's output is unknown, so whether an attribution is faithful cannot be checked. We present DEGRADX, to our knowledge the first testbed that supplies attribution ground truth at every observation for a regression target under sustained mean-nonstationarity. Its generator is fitted to three public battery-degradation datasets, MATR, HUST and FORM. Each fitted profile is evaluated against its dataset by statistical comparison of the fitted properties and the cross-channel correlation, by a discriminator between generated and measured sequences, and by the Train on Synthetic, Test on Real (TSTR) procedure. Under TSTR, a model trained on the HUST profile reaches 1.17 [0.94, 1.34] times the error of a model trained on measured cells, and the MATR and FORM profiles 1.49 and 1.54 times. Trained models learn the target and use every channel that enters it. The one localized effect in the data, a capacity recovery in FORM, is too small for trained models to be shown to learn it. Tests of the evaluation scores on controlled degradations of the ground truth show that rank agreement barely changes when an attribution map is smoothed or partly shifted in time. A temporal profile error introduced here registers both failures. On a reference model whose exact attribution is known, Integrated Gradients, feature occlusion and TimeSHAP recover it. On trained models, however, scores vary more across ten equally accurate models than between the methods in all but one case, indicating that they depend more on which model was trained than on the method. In practice, DEGRADX can be used to check whether an attribution method recovers a known attribution before the method is applied to models of degrading systems. It also shows which failures of an attribution map a given evaluation score registers.

Keywords Explainable artificial intelligence · Feature attribution · Ground-truth evaluation · Time-series regression · Prognostics and health management · Synthetic data generation

1 Introduction

Industrial assets degrade in service: bearings wear, rotating machinery loses efficiency, and battery cells lose capacity. Condition monitoring records this progression, directly or indirectly, in sensor time series, and maintenance policies that act on it are economically consequential. A survey of United States manufacturers estimates the total annual cost of inadequate maintenance at $222.0 billion and reports that respondents relying more heavily on predictive than on preventive maintenance experienced 18.5 % less unplanned downtime and 87.3 % fewer defects [1]. Data-driven models support the tasks on which such policies depend: detecting incipient faults, estimating the current health state, and predicting the remaining useful life (RUL). Because the resulting decisions are consequential, a numerical estimate is seldom sufficient on its own. Before a model is relied upon, engineers need evidence that it responds to the physical signature of degradation rather than to the incidental structure introduced by the acquisition process or by the

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composition of the training set, and an estimate that triggers an intervention must be justifiable to those who act on it. Post-hoc attribution is by far the most frequently applied method across manufacturing and industrial cyber-physical systems, as it can be applied to a model that is already trained and operational without altering it [2].

Attribution methods assign a value to each element of an input sequence, representing its contribution to the model's output. Whether such a method is faithful (that is, whether the returned values reflect the model's actual dependence on the input) is a separate question from predictive accuracy: a model may estimate its target well while the attribution computed for it misidentifies which inputs drove the estimate. Evaluation of faithfulness requires attribution ground truth specifying the actual contribution of each input element, and on measured data, no such ground truth exists. Validation therefore proceeds by perturbing input elements in order of estimated importance and measuring the effect on model performance. However, the reliability of this procedure is contested at several levels. Occluding elements skew the input distribution, so the model is queried on samples unlike those it was trained on, which is among the drawbacks that motivated dedicated evaluation frameworks for time series [3]. Still, such frameworks have themselves since been shown to contain significant weaknesses [4]. The metric commonly used in region perturbation, the Area Under Perturbation Curve (AUPC), was shown to be inadequate on its own for estimating faithfulness and can support incorrect conclusions, revealing neither how well relevant and irrelevant elements are separated nor how consistently, with results depending materially on the perturbation method applied and region size selection [5]. Under controlled conditions with known relevant elements, the disagreement is systematic, with perturbation-based and ground-truth-based metrics ranking the same attributions in opposite orders [6]. Such measures can therefore establish that two methods differ, but not which of them is closer to the model's actual dependence structure.

The established response is to construct data for which the attribution ground truth is known. This design underlies the synthetic evaluation benchmarks used throughout the time-series explainability literature [7, 8, 9]. However, degradation data differs from the processes these benchmarks generate in three respects. *First*, the target is continuous rather than categorical. *Second*, the signal exhibits sustained mean-nonstationarity, and this component carries the larger part of the target-relevant information. *Third*, the mean component contributes to every observation rather than to a few localized patterns alone, so relevance is graded rather than binary and is not exhausted by a set of positions. Established generators are built on the opposite premise: class-discriminating patterns are planted at known positions, and those positions constitute the attribution ground truth—an approach now standard enough to have its own package [10]. Neither the data nor this form of attribution ground truth transfers to prognostics, where attribution evaluation is therefore confined to functionally grounded indicators of the kind examined above [11].

This work presents DEGRADX, a testbed for evaluating attribution on multivariate time series whose regression target depends on a sustained nonstationary mean—the regime characteristic of degrading systems. It consists of a generator, a set of calibrated configurations, and an evaluation protocol. The generator produces multivariate sequences by adding a nonstationary mean component, localized patterns inserted at known positions, and noise. The target variable is an explicit additive function of the first two terms, so the contribution of every observation is known by construction rather than estimated. This yields attribution ground truth of two kinds: a dense graded field from the mean component, which contributes at every position, and a sparse set of positions carrying the inserted patterns. The protocol measures each on its own terms: rank agreement, a channel allocation error, and a temporal profile error, and precision-recall retrieval metrics standard in the literature [12, 9] against the sparse set. The generator parameters are fitted to public degradation datasets **MATR** [13], **HUST** [14], and **FORM** [15], each fitted configuration being called a *profile*. Synthetic evaluation invites the objection that conclusions drawn on generated data need not hold on measured data, and existing benchmarks acknowledge this gap rather than measure it, treating realism and ground-truth availability as a trade-off [16]. We instead report it: every profile is validated against the dataset it was fitted to, by distributional comparison and by the "Train on Synthetic, Test on Real" (TSTR) framework, so that each score is accompanied by evidence of how closely the profile reproduces the process it stands for. We therefore investigate three properties of DEGRADX.

- **RQ1:** How closely does each profile reproduce the dataset it was fitted to, in distribution and in task performance, and which properties resist calibration?
- **RQ2:** Is the resulting attribution ground truth usable, that is, learnable by a trained model, and does such a model use each channel that enters the target?
- **RQ3:** Do the scores discriminate between better and worse attribution maps, and which failures of a map does each of them register?

The proposed testbed is intended for any setting in which attribution is applied to a regression target under sustained mean-nonstationarity. Within prognostics, it supplies the attribution ground truth that evaluation currently lacks, allowing a method to be assessed before it is applied to a model. Beyond it, the testbed extends the evaluation surface of perturbation-based explainers, which are at present assessed almost exclusively on classification tasks over stationary

or state-switching processes. The generator, fitted parameters, seeds, trained models, ground-truth files, and scoring code are publicly released [2].

2 Related Work

The construction of synthetic data with known attribution ground truth is well developed for classification. Ismail *et al.* [7] built 70 datasets from seven base processes, including Gaussian noise, autoregressive variants, a Gaussian-process mixture, harmonic functions, and NARMA, marking informative features by an offset applied to one class, and evaluated saliency methods across architectures. Tonekaboni *et al.* [8] introduced the "feature importance in time" (FIT) framework, which was subsequently reused across the perturbation-based explainability literature [12, 17, 18]. Queen *et al.* [9] designed four synthetic datasets on a comparable base process and evaluated them together with four measured ones. Packaged environments have followed, covering perturbation-, gradient-, and example-based explainers [19], distributing pretrained models with precomputed explanations [20], and supplying the planting procedure itself as reusable software [10]. In every case, the attribution ground truth is defined by class-discriminating structure, an offset applied to one class, or a transient that determines the label. The same construction has more recently been turned on the evaluation procedure itself, using planted data to establish how perturbation-based and ground-truth-based scores diverge [6], and the authors of the most recent of these resources state that it is limited to classification, listing forecasting among the extensions left to future work [20].

A second group of benchmarks provides ground truth that describes the generating process rather than an individual prediction and therefore cannot be used to score an attribution map. Barić *et al.* [21] specified synthetic datasets whose generating process describes interactions between multiple series at increasing complexity and reported that models with satisfactory predictive performance frequently yield incorrect interpretations—an observation consistent with the separation between accuracy and faithfulness set out above. Their ground truth, however, is the interaction structure itself: which source series and which lags drive the target. TimeGraph [22] identifies the same neglect of nonstationarity that motivates this work, noting that trends, seasonality, irregular sampling, and unobserved confounders are generally absent from synthetic benchmarks, and supplies datasets that incorporate them. Its dynamics are therefore appropriate, but its ground truth is again structural—a causal graph over series and lags.

Attribution ground truth for regression is recent and takes two forms. *First*, series carrying known attribution ground truth are generated and then added to measured datasets so that the resulting semi-synthetic data retains real-world distributions. Kamarthi *et al.* [23] apply this to hierarchical demand forecasting, adding generated series to four measured datasets and evaluating attribution from the resulting variation in the outputs. Attribution is nonetheless scored against the attribution ground truth of the generated series, which are built on a noise background and are surrounded but not altered by the measured datasets. Therefore, the signal being scored exhibits no sustained mean-nonstationarity. *Second*, the attribution ground truth is computed rather than planted. When the target variable is not an additive function of its inputs, it cannot be decomposed into one contribution per input, and an attribution method must instead be applied to the generating process itself, whose output is then taken as the attribution ground truth. This has been done for forecasting series carrying seasonal structure and external inputs [24]. The quantity recovered is the importance of each input variable across the dataset rather than the contribution of individual observations. Where no attribution ground truth is available, evaluation falls back on functionally grounded indicators, that is, criteria computed from model behavior alone. A recent survey of explainable AI for industrial RUL prediction builds its entire evaluation framework from indicators of this kind [11], and the same practice is adopted by benchmark resources whose measured datasets carry no relevance annotation [20]. The reliability of this practice is disputed. Under controlled conditions with known relevant elements, indicators of this kind and ground-truth-based measures rank the same attributions in opposite orders [6], and work on benchmark fidelity attributes part of the disagreement to confounding factors in the evaluation setup itself [25].

A separate group of works produces realistic synthetic sequences that closely resemble measured data without providing any attribution ground truth. Diffusion-based [26], generative-adversarial [27], and hybrid [28] models generate capacity-fade curves fitted to public datasets and report that data of this kind can replace a substantial fraction of laboratory measurements in prediction tasks. In each case, the generating function is learned rather than written down, so no explicit mapping from individual observations to the target variable exists, and their contributions cannot be recovered. These models therefore occupy the position complementary to the measured data, but not as a ground-truth benchmark for attribution methods. This literature has, however, established how the distance between generated and measured data is estimated. Esteban *et al.* [29] introduced the TSTR procedure, training a model on generated data and reporting its error on measured data, which tests whether the dependence a model has to learn was preserved rather than only the appearance of the sequences. Yoon *et al.* [30] paired it with the classification error of a discriminator trained

[2] `[PLACEHOLDER: repository and DOI]`

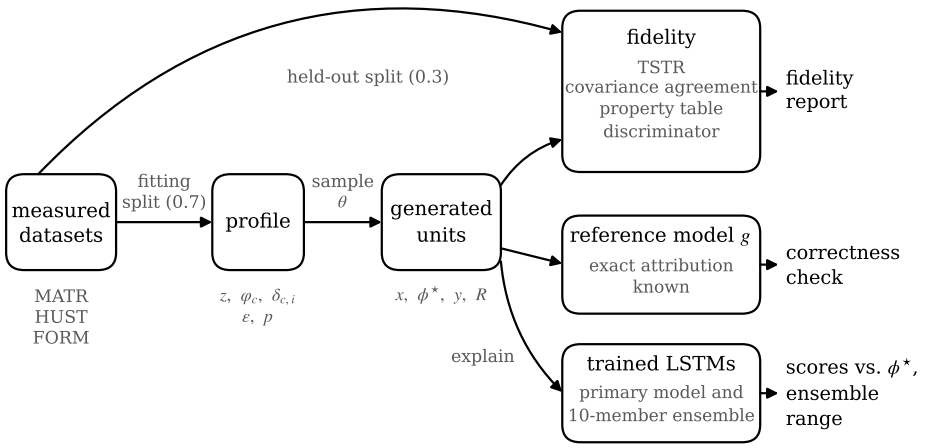


Figure 1: Structure of the testbed. Generator parameters are estimated from the fitting split of a measured dataset (MATR, HUST or FORM; 0.7 of its units, the remaining 0.3 held out), giving a profile: the degradation state z, the channel mappings φ_c, per-unit offset $\delta_{c,i}$, the noise ε, and the inserted patterns p. Drawing θ per unit generates units, each carrying the sequence x, the attribution ground truth $\phi^\star$, the decomposable target y, and the transfer target R. The generated units are compared with the held-out split to assess the fidelity of the profile. Attribution methods then explain the generated units on two kinds of model: the reference model g, whose exact attribution is known, so that it checks the correctness of each method; and trained LSTMs, a primary model and an ensemble of equally accurate models, whose attributions are scored against $\phi^\star$, with the range of the ensemble, against which differences between methods are read.

to separate the two, and Jeha *et al.* [31] added a Fréchet-style distance computed in a learned representation space. Measures of this kind are reported by the degradation generators above [26], and by *none* of the attribution benchmarks of the preceding paragraphs. The authors concede the gap between generated and measured data and treat realism and ground-truth availability as a trade-off, given that the generation process for measured data is unknown [16].

Across the overviewed body of work, the properties DEGRADX requires are available separately and never together. The synthetic benchmarks supply exact attribution ground truth, but for a categorical target and in the form of planted positions, which a component present at every observation cannot be reduced to. The regression benchmarks provide either attribution ground truth for a signal with no sustained nonstationarity or ground truth describing the generating process rather than individual observations. Where neither is available, the evaluation proceeds with no reference at all. The generators of degradation data supply both the signal and the measured correspondence to it, but their generating function is learned, and the contribution of an observation cannot be read off it. DEGRADX occupies the position that remains. It supplies attribution ground truth defined at every observation for a continuous target under sustained mean-nonstationarity, and it applies the measures of the generative literature to an attribution testbed: each profile is compared with the dataset it was fitted to in distribution and under TSTR by whether a model trained on the profile predicts on that dataset. The second is the more demanding of the two, since it tests the dependence between the sequences and the target variable rather than the sequences alone. How closely a profile matches on both is reported with every score it produces. The match bears on the sequences on which attribution methods are tested and on their relation to the RUL: the closer the match, the stronger the case that methods are scored on sequences like those of the measured process—a case that existing benchmarks, by the trade-off conceded above, cannot make.

3 Methodology

DEGRADX consists of a generator, a fitting procedure that adapts the generator to a measured dataset, and an evaluation protocol. This section describes them in that order. Figure 1 shows how the parts connect.

3.1 Generator

A generated unit consists of a multivariate sequence $x \in \mathbb{R}^{T\times C}$, where $t \in \{1,\dots,T\}$ indexes positions and $c \in \{1,\dots,C\}$ indexes channels. In the datasets used here one position corresponds to one charge–discharge cycle,

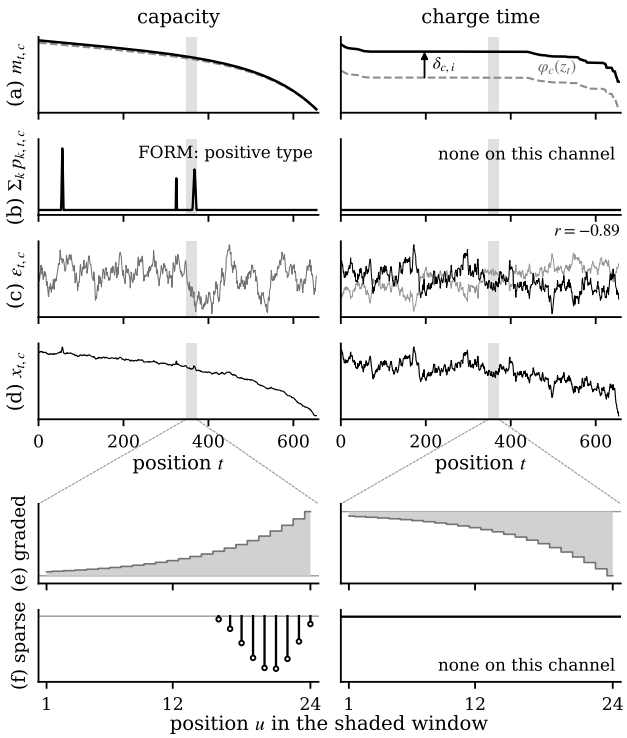


Figure 2: Two channels, capacity (left) and charge time (right), of a generated FORM unit: (a) mean component, with its shared part $\varphi_c(z_t)$ dashed; on charge time the two differ by the unit's offset $\delta_{c,i}$, (b) inserted patterns, here positive capacity recoveries inserted at uniformly drawn positions, (c) noise, standardised; the noise of mean discharge voltage (grey) is correlated with that of charge time, $r = -0.89$, (d) their sum, which is what a model receives, and the resulting attribution ground truth $\phi^\star$ on the shaded window: (e) the graded field from the mean component and (f) the sparse set from the inserted pattern that falls in the window. Both channels fall with degradation, so their weights are negative: a capacity recovery lowers the target, and on charge time the unit's positive offset makes the graded field negative. Vertical scales are arbitrary.

and T is the position at which the unit reaches its end of life (EOL). Each channel is the sum of three terms,

$$x_{t,c} \;=\; m_{t,c} \;+\; \sum_k p_{k,t,c} \;+\; \varepsilon_{t,c}, \tag{1}$$

where m is a mean component, $p_{k,t,c}$ is a set of inserted patterns, and ε is noise. Figure 2 shows the three terms and their sum for two channels of one unit. This composition (a base process into which patterns are inserted at recorded positions) is the design used by established synthetic benchmarks [8, 9]. What differs is the role of the first term. There the base process is background, and the attribution ground truth does not depend on it. Here the mean component carries the degradation trajectory and enters the target variable, which is what makes it graded rather than confined to the inserted patterns.

3.1.1 Mean Component and Degradation State

Degradation is one underlying process observed through several measured quantities, and those quantities do not follow the same shape: e.g., capacity falls while internal resistance rises. Fitting one parametric curve to every channel would impose a shape that only some of them have. Fitting an independent curve to each channel would instead leave the relationship between channels unconstrained. The generator therefore separates *how far a unit has degraded* from *what that degree of degradation looks like in a given channel*. The first is a scalar degradation state z_t, the second a mapping φ_c per channel, to which each unit i adds a constant offset $\delta_{c,i}$,

$$m_{t,c} \;=\; \varphi_c(z_t) \;+\; \delta_{c,i}. \tag{2}$$

The mappings φ_c are monotone, continuously differentiable functions of the degradation state, shared across all units of a profile, and are fitted from the measured data. The offset $\delta_{c,i}$ shifts unit i in channel c by the same amount at every

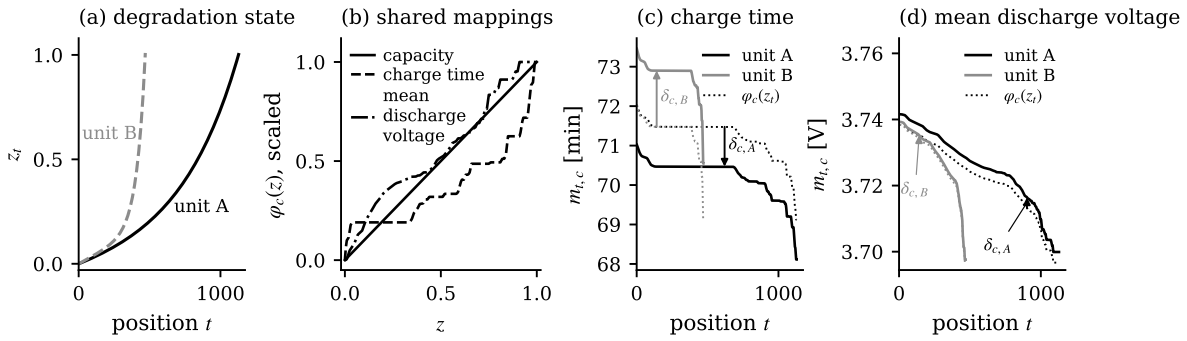


Figure 3: Construction of the mean component, on the FORM profile. Two units advance through the degradation state at different paces and reach EOL at different positions (a). Fitted mappings (b), scaled to their change from $z_t = 0$ to EOL, are applied to both. In charge time (c) and mean discharge voltage (d), each unit's mean component is the shared mapping at its own state (dotted) shifted by the unit's offset $\delta_{c,i}$ (solid).

position. It is fitted from the measured data jointly with the mappings, and a generated unit takes the offsets of the measured unit it is drawn from. Measured cells differ in these levels because of manufacturing variation and of the protocols under which they are formed and cycled. The capacity channel carries no offset. Units therefore differ in the pace at which they age and in their levels, but not in what a given degree of ageing looks like. Figure 3 illustrates this for two units with different lifetimes. Each profile also contains two null channels, which do not depend on the degradation state. The first is a constant plus noise with the autocorrelation of the capacity noise, so that it cannot be told apart by its smoothness. The second holds the measured values of a channel permuted across units and positions, so that it cannot be told apart by its values. Since a model has no reason to use them, any attribution they receive is error of the attribution method. They give every profile channels of zero target weight, on which this error is measured.

Definition of the degradation state and of EOL. Let q_t denote the capacity channel, smoothed by a declared filter, let q_{nom} be the nominal capacity documented for the dataset, and let q_1 be the maximum of q_t over the first $k := 20$ positions, attained at position t_1. EOL is reached when capacity has fallen to a fraction ρ of nominal capacity, and the degradation state measures how much of the fall from q_1 to that threshold has been consumed,

$$z_t \;=\; \frac{q_1 - q_t}{q_1 - \rho\, q_{\mathrm{nom}}}, \qquad T \;=\; \min\{\, t \geq t_1 : z_t \geq 1 \,\}. \tag{3}$$

The degradation state runs from 0 at the early-life reference to 1 at EOL. Because both endpoints are fixed in advance ($\rho\, q_{\mathrm{nom}}$ by design and q_1 by observation), z_t is a transform of the current capacity alone, and is computable from data available up to t. The more obvious alternative—rescaling each unit's trajectory to the unit interval over its observed length—would make z_t a function of t/T, encoding the unit's total lifetime at every position in every channel. Since the total lifetime determines the RUL, the generated data would exhibit target leakage. A model trained on it would depend on information that is unavailable at prediction time, a failure mode reported for RUL pipelines [32].

Measured datasets do not agree on ρ. The common convention is a fall to 80% of nominal capacity. However, the experiments underlying some datasets were stopped at a fade of 20 to 30%, and individual units within a dataset were carried to different final capacities. Adopting each dataset's own convention would define z_t against a different endpoint in each profile and would render the parameter distributions fitted from them mutually incomparable. We therefore apply a single $\rho := 0.8$ across all profiles, taken against the nominal capacity each dataset documents. A unit whose early-life reference lies within $0.05\, q_{\mathrm{nom}}$ of the threshold is excluded. Its capacity can fall by less than 5% of nominal capacity before reaching EOL, which is too small to define the degradation state. Some datasets were recorded only until capacity reached the threshold. A unit, measured or generated, whose last recorded capacity lies within $0.01\, q_{\mathrm{nom}}$ of the threshold is therefore taken to reach EOL at its last position. Units that never reach the threshold have no EOL and therefore no RUL. They are excluded from the quantities that depend on EOL, namely the choice of the parametric family, the distribution of θ, the distribution of unit lengths and the transfer measurement. They are retained for the others: the mappings, the offsets, the noise and the inserted patterns.

Shape of the degradation state. The trajectory q_t, and hence z_t, follows a parametric curve with a low-dimensional parameter vector θ drawn per unit from a distribution estimated from measured units. The functional form is not fixed in advance. Three empirical fade models are fitted to each dataset: the two-term exponential established for capacity fade [33], a rollover form whose parameters are the initial rate, the transition position, the transition width, and the

terminal rate [34], and a power law. The family is selected per dataset by its cross-validated fit error across units, so that the shape carried by a profile is decided by the data it stands for rather than by us.

3.1.2 Inserted Patterns and Noise

Inserted patterns. Measured degradation trajectories are not smooth. Alongside the global degradation trend, capacity sequences carry local fluctuations. Established work separates the two, decomposing the trajectory into a degradation trend, a regeneration component, and a residual fluctuation component [35]. The positive fluctuations are *capacity regeneration*: a temporary recovery of measured capacity that follows a rest period between cycles and persists over the cycles after it [36, 37]. Negative fluctuations follow from the conditions under which a cycle is measured rather than from the state of health of the cell. Ambient temperature is the documented instance, appearing as chamber temperature excursions in public cycling datasets [13]. In both cases the deviation is treated as separate from the degradation trend, which is estimated and extrapolated in its own right [35, 36], and not as a change in how far the unit has degraded. In this work, these fluctuations are incorporated, since profiles without them would be smoother than the dataset they stand for. They also supply the localized part of the attribution ground truth, which is the part the retrieval metrics reported on existing benchmarks are defined on. The generator represents these effects as inserted patterns $p_{k,t,c}$, each described by a position, a signed amplitude, and a duration. Because they act on the observation and not on the degradation state, they do not alter z_t or the position of EOL. A profile enables only the inserted pattern types its dataset exhibits.

Noise. Noise is added per channel with variance and autocorrelation set during fitting, and has zero mean. Within a channel it is a first-order autoregressive process driven by a white-noise term at each position. Since the residuals of measured channels move together within a unit, the white-noise terms of the channels of a unit are drawn jointly. Their correlation matrix, one per profile, is chosen so that the noise correlation between two channels equals the within-unit correlation of their measured residuals after the offsets are removed. The null channels stay independent of every other channel.

3.2 Targets and Attribution Ground Truth

Two quantities have to be distinguished. A *target* is a value a model is trained to predict from a window of the sequence. The *attribution ground truth* is the contribution that each position and channel of that window makes to the target. It is the quantity an attribution method is asked to recover and against which its output is scored. The generator defines two targets, for two different purposes. The *decomposable target* y is written down as an additive function of the generated components, so that its attribution ground truth follows from its form. It is the target used whenever attributions are scored. The *transfer target* R is the RUL, has a counterpart in every measured dataset, and is used only to test how well a profile reproduces the dataset it was fitted to.

3.2.1 Decomposable Target and Attribution Ground Truth

Let W be a window of $L := 24$ consecutive positions ending at position t, and let u index the positions within it, $u \in \{t - L + 1, \dots, t\}$. The decomposable target on that window is a weighted sum of the generated components:

$$y \;=\; \sum_{(u,c)\,\in\,W} w_{u,c}\left(m_{u,c} - x_c^0\right) \;+\; \sum_{(u,c)\,\in\,W} w_{u,c} \sum_k p_{k,u,c}. \tag{4}$$

Since $m_{u,c}$ is $\varphi_c(z_u)$ up to the unit's levels (Equation 2), the first term is a fixed function of the degradation state over the window. The target y thus summarises how far the unit has degraded across those positions, perturbed by whatever transients occurred in them. It is important to note that equation 4 *is not a physical law and is not proposed as a model of how RUL arises from a degradation trajectory*. The decomposable target is never compared against measured data, whereas the sequences and the transfer target are. Because Equation 4 is additive and written down rather than learned, the contribution of every position and channel is known exactly:

$$\phi^{\star}_{u,c} \;=\; w_{u,c}\left(m_{u,c} - x_c^0\right) \;+\; w_{u,c} \sum_k p_{k,u,c}. \tag{5}$$

The first term of Equation 5 is the graded field and the second the sparse set, both measured from the reference point $x_c^0 := \varphi_c(0)$. The weight $w_{u,c} = \kappa_c\,\pi_u$ is the product of a channel factor and a position profile. The channel factor $\kappa_c = \beta_c/(\varphi_c(1) - \varphi_c(0))$ makes the full change of each of capacity, charge time and mean discharge voltage between the pristine level and EOL contribute $\beta_c := 1/3$ to the target. It is zero on every other channel. Since the graded field outweighs the patterns on the channel that carries them, every generated window is supplied with its counterpart without inserted patterns. A method's attribution on the counterpart is scored against the graded field, and the difference

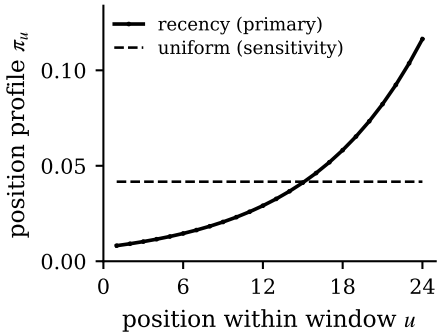


Figure 4: Weighting of positions within a window of $L := 24$ positions. The recency weighting keeps every position relevant while giving the graded attribution ground truth an ordering; the uniform weighting, reported as a sensitivity, gives it none.

between its attributions on the window and on the counterpart against the sparse set. The position profile π_u decides how relevance is distributed across the positions of a window, and therefore what shape a correct attribution map has.

Figure 4 shows the two weightings. *Recency* weighting, $\pi_u \propto 2^{-(t-u)/h}$ normalized to sum to one over the window, increases smoothly toward the end of the window and halves every h positions toward its start. With the half-life $h := L/4 = 6$, the window is four half-lives long and its first position keeps $2^{-(L-1)/h} \approx 0.07$ of the weight of its last. Every position thus stays relevant while the field has an ordering a method can be scored against. *Uniform* weighting, $\pi_u = 1/L$, makes every position equally relevant. It is the limit in which no ordering exists to recover. Recency weighting is primary, and uniform weighting is a sensitivity readout. Reporting both therefore measures how far a result depends on where relevance is placed rather than on which method produced it.

Reference model. Attribution is evaluated on a trained model, so a low score may reflect error introduced by the attribution method or error in the model itself. To separate the two, the testbed also provides a *reference model*, the affine map

$$g(x) \;=\; \sum_{(u,c)\,\in\,W} w_{u,c}\left(x_{u,c} - x_c^0\right), \tag{6}$$

which applies the weights w to the deviation of the observed window from the reference point. It reproduces the target exactly when noise is absent and otherwise differs from it only through $\sum_{(u,c)\in W} w_{u,c}\,\varepsilon_{u,c}$, whose expectation is zero. Because the map is affine and its weights are known, attributions computed for it can be compared against an analytically available solution. Any remaining error is then attributable to the attribution method alone. The reference model establishes what an attribution method achieves when the model is exactly right, and says nothing about whether the target is well chosen.

3.2.2 Transfer Target

The decomposable target has no counterpart in a measured dataset, so a model trained to predict it cannot be evaluated on measured data. The generator therefore records a second label, the RUL at position t,

$$R_t \;=\; T - t, \tag{7}$$

with T as defined in Equation 3. This is the same quantity that measured datasets provide as a RUL label. Two points follow from Equation 7. First, R is not a function of z_t: two units that have reached the same degradation state have different RUL according to how fast they are advancing through it. Second, R depends on the whole trajectory and is therefore a label and never an input. Neither R, nor T, nor any quantity derived from them is available to a model at prediction time.

3.3 Fitting the Generator to a Measured Dataset

A *profile* is the set of generator parameters obtained from one measured dataset. Fitting uses the fitting split only, so that the units later used to assess the profile take no part in producing it. The procedure has six steps.

First, the capacity channel is extracted for every unit, and readings that no degradation process produces are removed. The series is smoothed by a Savitzky–Golay filter with a window of 11 positions and polynomial order 2, and EOL

is located by Equation 3. *Second*, the three parametric families are fitted to each unit's capacity trajectory from the early-life reference to EOL by nonlinear least squares, and one of them is selected for the dataset. The empirical distribution of the fitted parameters over units is the profile's distribution of θ. A generated unit resamples one fitting unit that reaches EOL and takes its θ together with its early-life reference, its offsets, its noise parameters, and its pattern rate, so that the dependence among them across units is kept. *Third*, the mappings φ_c of Equation 2 are fitted as smooth monotone functions of the fitted degradation state, jointly with the offsets $\delta_{c,i}$, and the covariance among channels is recorded. Mappings and offsets are fitted by backfitting, which alternates a fit of the mapping to the readings less the offsets with an update of the offsets, until neither changes by more than 10^{-4} of the channel's range $|\varphi_c(1) - \varphi_c(0)|$. Each offset is updated to the median over the unit's positions of its readings less the mapping, centred to median zero over units. For the capacity channel the mapping is affine by construction, since Equation 3 defines the degradation state as an affine transform of that channel. *Fourth*, the residuals of the trajectory fit, and in the other channels the residuals about the mapping plus the unit's offset, supply the noise variance and autocorrelation. Their correlation between channels, estimated within each unit and averaged over units, sets the correlation of the noise across channels. *Fifth*, the same residuals supply the inserted patterns, by the detection procedure stated below. *Sixth*, the empirical distribution of unit lengths is recorded.

The distinction between what is measured and what is chosen is reported explicitly, since it determines what the testbed can claim. Estimated from the measured dataset are the distribution of θ, the choice of family, the mappings φ_c, the per-unit offsets $\delta_{c,i}$ and early-life references, and the covariance among channels. Also estimated are the noise variance and autocorrelation together with the within-unit correlation of the noise between channels, the rate, amplitude, and duration of the inserted patterns, and the length distribution. Declared by design are the EOL fraction ρ, the nominal capacity of each dataset, the number k of positions over which the early-life reference is taken, the fraction of nominal capacity beyond which a single reading is removed, and the tolerance within which a truncated record reaches EOL. Also declared are the tolerance of the offset fit, the smoothing window length and polynomial order, the pattern detection thresholds, the channel shares β_c and position profiles of the weights w, and the window length L.

Detecting the inserted patterns and estimating their parameters. Two families of detection procedure are established in battery prognostics. The first is triggered by the capacity sequence itself: it tests the residual of a fitted degradation trend for a departure the trend does not account for, for instance by a statistical test on the residual sequence [38]. The second is triggered by the rest time between adjacent cycles, on the mechanism that a rest longer than a threshold produces a regeneration over the cycles that follow [36, 37]. We adopt the first. Rest time is recoverable from the timestamps of some public datasets and not of others, whereas the residual of the trajectory fit is available for every dataset by step two above. The same rule has to apply to every profile for the fitted parameters to be comparable across them.

The rule is as follows. A candidate pattern is a run of consecutive positions whose residual from the fitted trajectory exceeds 2.5 times the residual scale and whose length lies between two positions and the 11 positions of the smoothing window. The sign of the run gives the type. For each detected pattern we record the position of its extremum, its signed amplitude, and its duration in positions. Over the dataset, we estimate the distribution of amplitude and of duration together with the per-unit rate of occurrence. Amplitude and duration are the quantities the rest-time literature also extracts from measured sequences, as the value reached at the regeneration cycles and the number of cycles the regeneration region spans [36]. The fitted distributions are therefore comparable with published characterizations of the same phenomenon [36, 37]. A type is enabled in a profile only if the rule detects it at least twice as often as in noise simulated with the profile's own fitted variance and autocorrelation. Detections that the fitted noise alone produces are the noise the threshold exists to exclude.

Splitting and use of future information. Windows drawn from one unit are strongly dependent, so all splits are performed at the level of units rather than windows, on both generated and measured data. Normalization statistics are computed on training units alone. No input feature may be derived from a quantity that depends on positions after t or on the full length of a sequence, since that length is the label. This restriction applies to model inputs. It does not apply to the fitting procedure above, which is not a model and is allowed to use each measured unit in full, nor to the targets, which are labels.

3.4 Profiles

Three profiles are built and evaluated. **MATR** [13] supplies the reference channel set, recording discharge capacity, internal resistance, charge time, mean discharge voltage, and temperature statistics per cycle. **HUST** [14] contains cells of the same chemistry and format cycled under different discharge protocols in a different laboratory. It therefore tests whether the fitting procedure transfers when the cell is held fixed and the protocol is not. The third profile, **FORM**, is built from the formation dataset of Cui *et al.* [15], whose cells differ from those of MATR and HUST in chemistry,

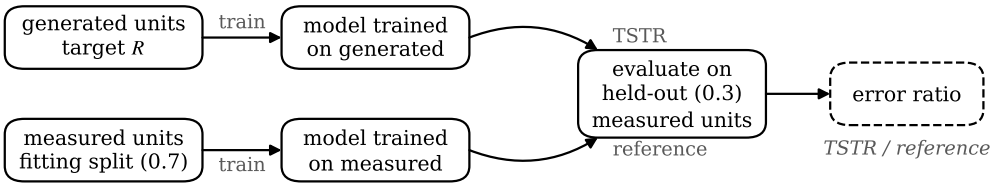


Figure 5: The transfer measurement. Two models of the same architecture are trained, one on a profile with the transfer target R and one on the measured units used to fit it, and both are evaluated on measured units held out from fitting, against their RUL label. The transfer ratio is the ratio of their errors.

format and laboratory. The cells were formed under different protocols and then aged under one common protocol, interrupted every 100 aging cycles by a diagnostic block that rests the cell and discharges it at a lower rate. Positions are the aging cycles, and the capacity channel is the discharge capacity of the aging cycle.

3.5 Evaluation Protocol

3.5.1 Profile Fidelity

RQ1 asks how closely each profile reproduces the dataset it was fitted to. Each quantity estimated by the fitting procedure is estimated on generated units and on the held-out measured units, and compared with its value on the fitting split. Two measures established in the generative time-series literature are computed on windows. The covariance agreement is the relative Frobenius distance between the channel correlation matrices of generated and held-out windows. It is compared with the same distance between the measured fitting and held-out windows, which shows how far two samples of the same measured dataset differ. The discriminative error is the classification error of a discriminator trained to separate generated from measured windows [30]. At 0.5 the two cannot be told apart. It is a coarse indicator only. On sequences of the lengths considered here, a discriminator separates the two sources whenever any property is mismatched, and its error does not indicate which property is responsible. The property-level comparisons therefore carry the evidence.

The transfer measurement follows the TSTR procedure [29] with the transfer target of Equation 7, and yields the transfer ratio of Figure 5. A ratio of one means that a model learns the dependence between the measured sequences and their RUL as well from the profile as from the measured units. A ratio above one means that the profile lacks part of that dependence. Each profile is evaluated against its own paired dataset only, since a pooled evaluation would allow a close match on one dataset to compensate for a poor match on another.

3.5.2 Usability of the Attribution Ground Truth

RQ2 asks whether the attribution ground truth is usable, which requires that a model can learn the decomposable target, that it uses each channel that enters the target, and, where a profile carries inserted patterns, that it learns the pattern term. Learnability is measured by the normalized root-mean-square error (NRMSE) of trained models on generated test units. Each of the two terms of Equation 4 is removed from the training target in turn and the model retrained. If its error on y does not increase, the model has not learned the removed term.

Since every channel other than the null channels carries the same degradation state, an accurate model need not read every channel to which $\phi^\star$ assigns part of the target. Where it does not, the channel-level ground truth describes a function the model did not learn. Each weighted channel is therefore permuted conditionally on the remaining channels, by replacing it with its ridge-regression prediction from them plus residuals permuted across test windows. A channel is taken as used if the 95% interval over test units of the resulting increase in error lies above zero. The null channels are permuted to check that a model does not use them.

3.5.3 Responsiveness of the Scores

RQ3 asks whether the scores distinguish better from worse attribution maps, and which failures of a map each of them registers. A map A is scored against the attribution ground truth in five ways. *Rank agreement* is the rank correlation between A and the graded field over the cells of a window. It depends only on the order of the cells. On a field that grows steadily toward the end of the window, smoothing a map along the window or moving part of each channel's mass to its last position leaves that order largely unchanged. Two further scores therefore measure separately how a map divides attribution among channels and how it distributes attribution along the window. The *channel allocation*

error is the total-variation distance between the shares of absolute attribution on each channel,

$$\frac{1}{2}\sum_{c}\left|s_c(A) - s_c(\phi^\star)\right|, \qquad s_c(M) \;=\; \frac{\sum_u |M_{u,c}|}{\sum_{u,c'} |M_{u,c'}|}, \tag{8}$$

taken over all channels, those of zero weight included. It is 0 when the shares agree and at most 1. The *temporal profile error* is computed for each weighted channel as the 1-Wasserstein distance over positions between $|A_{\cdot,c}|$ and $|\phi^\star_{\cdot,c}|$, each normalized to unit mass (for positions one apart, the sum over the window of the absolute difference between their cumulative shares). It is averaged over the weighted channels with weights equal to the mass of $|\phi^\star|$ on each. It is measured in positions, is 0 when the profiles agree, and is at most $L-1$. The remaining two scores are the share of absolute attribution on zero-weight channels and, where inserted patterns exist, retrieval of the sparse set.

RQ3 is answered without involving any attribution method. The ground-truth field is a correct attribution map by construction, so it can be degraded in controlled amounts. It is degraded by adding noise of increasing variance, shifting a known fraction of each channel's mass toward the start or the end of the window, smoothing along the window, and permuting a known fraction of the cells. A score that does not change as the map is degraded does not register that failure. The change of each score is expressed as a fraction of the distance between its value on the undegraded field and its value on a fully permuted map, which is its chance level.

3.6 Models and Baselines

Attribution is evaluated on a single trained architecture, a recurrent network with long short-term memory (LSTM) units, together with the reference model. The testbed scores attribution maps against known contributions independently of the model that produced them, so a second trained architecture would not contribute to the RQs. On the reference model, the *exact attribution*, $w_{u,c}\,(x_{u,c} - x^0_c) = \phi^\star_{u,c} + w_{u,c}\,\varepsilon_{u,c}$, differs from $\phi^\star$ only by the weighted noise of the window. Its scores set the ceiling of the scores on that model. A trained model, by contrast, need not learn the function that defines $\phi^\star$. The channels of a window share one degradation state, so other functions reproduce the target equally well, and its scores also reflect which function it learned. To bound this effect, ten equally accurate LSTMs, of two sizes with five initializations each, are trained. Each method's scores on one of them, the *primary model*, are reported with their range over all ten.

Three published methods are run at their default configurations, apart from the baseline, on both the trained and the reference model: TimeSHAP [39] for perturbation-based Shapley attribution over sequences, Integrated Gradients (IG) [40] for gradient-based attribution, and feature occlusion [41]. Methods that optimise a mask per instance [12, 17, 18] are excluded, since their scores depend on hyperparameters whose tuning is a comparison of methods rather than a reference measurement. The scores are reported in aggregate, as reference values for later work rather than as an evaluation of the methods themselves.

Each method explains a window relative to a *baseline*, an input of the same shape as the window. IG integrates gradients along the path from the baseline to the window, feature occlusion replaces occluded cells by their baseline values, and TimeSHAP takes the values of cells excluded from a coalition from the baseline, its background instance. Shapley- and path-based attributions sum to the difference between the model output on the window and on the baseline, so the baseline fixes the point from which contributions are measured. Since $\phi^\star$ is measured from the reference point x^0, the three methods share one baseline, the pristine window, every position of which is at x^0. Unlike a zero input, it lies close to the first windows of every unit, so the trained model is not queried on a window unlike any it was trained on. To show whether the reference values depend on this choice, TimeSHAP is also run from the average event it uses by convention. It is then scored against the ground truth for that baseline, which is Equation 5 with x^0_c replaced by the average event.

4 Results

Intervals are 95% bootstrap intervals over units. MATR rests on the 135 cells of the three batches of [13], of which 128 reach EOL, HUST on all 77 cells of [14], and FORM on all 182 aged cells of [15]. Every cell of HUST and FORM reaches EOL. The fitting splits hold 95, 54 and 127 cells and the held-out splits 40, 23 and 55. Models trained on the decomposable target use the 300 units generated for each profile at one generation seed: 210 for training, 45 for validation and 45 for testing.

The profiles differ in whether they carry inserted patterns. On the fitting splits of MATR and HUST, capacity-residual runs occur at most at the rate at which the same rule detects runs in fitted noise. Neither profile therefore carries inserted patterns, and the graded field is its whole attribution ground truth. In FORM the positive runs occur at 2.28 times that rate, with a median amplitude of 2.7 mAh, and the type is enabled; the negative runs are not.

Table 1: Property-level comparison between each profile and the dataset it was fitted to, each value estimated by the same fitting procedure; medians where a distribution is summarized. Residual correlation: relative Frobenius distance of the within-unit residual correlation matrix from that of the fitting split, and the mean absolute difference of its entries over the pairs that include capacity and over the other pairs.

	MATR			HUST			FORM		
Property	fitting	held-out	gen.	fitting	held-out	gen.	fitting	held-out	gen.
Fit error of the profile's family [fraction of q_1]	0.0061	0.0054	0.0033	0.0027	0.0032	0.0014	0.0105	0.0109	0.0077
Drift rate [mAh/100 cyc.]	25.6	25.6	23.9	16.8	16.3	17.0	7.78	8.27	7.87
Transition position [$\times T$]	0.999	0.999	0.999	0.686	0.651	0.676	0.999	0.998	0.999
Unit length T [cycles]	770	744	823	1860	1938	1807	701	665	683
Early-life reference q_1, IQR [mAh]	10.8	10.4	12.4	21.2	19.6	26.9	2.63	2.31	3.43
Noise variance, capacity [mAh^2]	2.82	4.57	1.88	1.10	1.06	0.945	1.43	1.56	0.989
Noise variance, charge time [min^2]	0.694	0.799	0.584	1.39	1.43	1.02	0.765	0.624	0.575
Noise autocorrelation, charge time	0.89	0.81	0.88	0.99	0.99	0.99	0.95	0.95	0.94
Offset IQR, charge time [min]	4.77	1.30	4.96	1.68	1.62	1.97	1.43	1.21	1.31
Offset IQR, mean discharge V [mV]	59.0	53.2	63.8	53.0	47.4	54.2	13.3	8.48	12.9
Positive residual runs per 100 positions	0.12	0.12	0.09	0.06	0.04	0.11	0.33	0.29	0.27
Residual correlation, rel. Frob. to fitting split	0.00	0.09	0.11	0.00	0.03	0.03	0.00	0.10	0.13
capacity pairs, mean $\lvert\Delta r\rvert$	0.000	0.026	0.075	0.000	0.023	0.029	0.000	0.065	0.109
other pairs, mean $\lvert\Delta r\rvert$	0.000	0.059	0.056	0.000	0.012	0.013	0.000	0.085	0.094

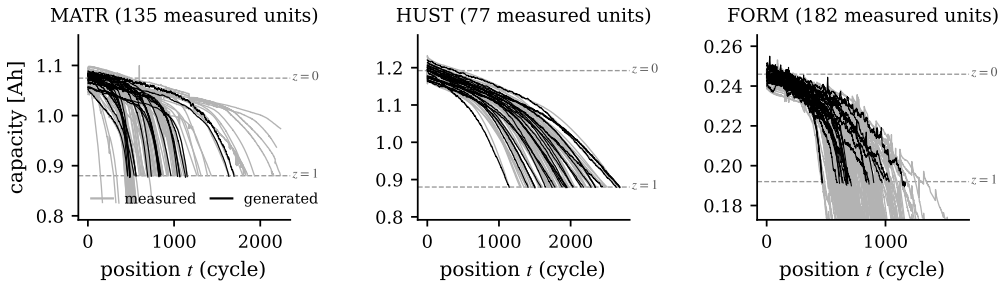


Figure 6: Capacity of the measured units (grey) and of 25 generated units (black) of each profile, with the pristine level $z = 0$ and the EOL threshold $z = 1$ of Equation 3 marked. Measured records are drawn to their end, which in FORM lies well past EOL; generated units end at EOL.

4.1 Profile Fidelity

Table 1 and Figure 6 show that the capacity trajectory is reproduced. In every profile, the median drift rate, the fraction of life at which the transition falls, and the median unit length of generated units are close to those of the fitting and held-out cells. The family fits generated units, which are drawn from it, with less error than measured ones. The spread of the offsets of the other channels is reproduced, and so is the within-unit residual correlation, apart from an excess in the capacity pairs of MATR and FORM. This excess comes from a bias of the estimator (Section 5.1). FORM's positive residual runs, inserted patterns and noise together, occur at 0.27 per 100 positions against 0.33 and 0.29 in the fitting and held-out cells. The departures concern spreads rather than levels. In every profile, the noise variances of generated units are lower than those of the fitting cells, and their early-life reference is more spread out.

In every profile, the distance between the channel correlation matrices of generated and held-out windows lies within the interval of the same distance between the measured fitting and held-out windows (Table 2). The generated windows thus differ from the held-out cells in correlation structure no more than two samples of the measured dataset differ from each other. In FORM, a discriminator separates generated from measured windows at every generation seed, with an error of 0.10 to 0.17, without indicating which property it detects. In MATR and HUST its error changes widely with the generation seed, so the discriminator does not establish whether, or how well, generated and measured windows can be separated.

The transfer ratio is 1.17 [0.94, 1.34] for HUST, 1.49 [1.04, 2.29] for MATR and 1.54 [1.28, 1.74] for FORM (Table 2). The interval of HUST contains one. On held-out measured cells, the error of a model trained on the HUST profile is therefore not distinguished from that of a model trained on the measured cells. Trained on the MATR or FORM profile alone, a model reaches about one and a half times the error of a model trained on the measured cells. Both intervals exclude one, so these two profiles reproduce part, but not all, of the dependence between the sequences and the RUL that the measured cells carry.

Table 2: Profile fidelity. Units: held-out measured units and generated units per generation seed. Discriminative error: range over three generation seeds; at 0.5 a discriminator cannot separate generated from measured windows. Covariance agreement: relative Frobenius distance between the channel correlation matrices of generated and held-out windows; in parentheses, the same distance between the measured fitting and held-out windows and its 95% interval. Transfer ratio: Figure 5.

Profile	Units	Discriminative err.	Covariance agr.	Transfer ratio [CI]
MATR	40 / 300	0.24–0.42	0.21 (0.17; 0.08–0.44)	1.49 [1.04, 2.29]
HUST	23 / 300	0.37–0.62	0.05 (0.04; 0.02–0.12)	1.17 [0.94, 1.34]
FORM	55 / 300	0.10–0.17	0.35 (0.30; 0.20–0.43)	1.54 [1.28, 1.74]

Table 3: Usability of the attribution ground truth under recency weighting, on the test units. Accuracy: NRMSE of the primary model and, in parentheses, the range over the ten models. Permutation importance and channel use: increase in the mean squared error normalized by the variance of y (NMSE) when a channel is permuted across windows or resampled conditionally on the other channels, with 95% intervals over test units; for the null channels, the upper end of that interval. –: no inserted pattern type is enabled in the profile.

Quantity	MATR	HUST	FORM
Accuracy, NRMSE of the primary model (ensemble range)	0.091 (0.088–0.096)	0.095 (0.089–0.097)	0.179 (0.176–0.190)
Offset term, share of Var(y)	0.128	0.343	0.159
Spearman correlation of y with RUL	-0.71	-0.61	-0.69
Permutation importance, capacity	0.908	0.525	0.876
Permutation importance, charge time	0.256	0.049	0.151
Permutation importance, mean discharge V	0.085	0.808	0.243
Permutation importance, null channels (upper bound, flat / permuted)	0.0001 / 0.0000	0.0000 / 0.0000	0.0003 / 0.0003
Channel use (conditional importance), capacity	0.076 [0.066, 0.093]	0.068 [0.057, 0.084]	0.146 [0.128, 0.181]
Channel use (conditional importance), charge time	0.053 [0.037, 0.092]	0.015 [0.012, 0.018]	0.094 [0.074, 0.120]
Channel use (conditional importance), mean discharge V	0.049 [0.038, 0.079]	0.630 [0.545, 0.744]	0.096 [0.076, 0.122]
Members using every weighted channel	10/10	10/10	10/10
Pattern term, variance ratio to the mean term	–	–	1.7×10^{-5}
Pattern term removed, NMSE increase	–	–	0.0000 [-0.0001, 0.0002]

4.2 Usability of the Attribution Ground Truth

The decomposable target is learnable on all three profiles (Table 3). On the test units, the primary model predicts it with an NRMSE of 0.091 in MATR, 0.095 in HUST and 0.179 in FORM, and the ten models are equally accurate within each profile. The offsets carry 0.128, 0.343 and 0.159 of the variance of y. In HUST, about a third of the variance of y thus records the level at which a cell sits rather than how far it has degraded. The rank correlation of y with RUL is -0.71, -0.61 and -0.69, so y grows as the RUL shrinks without being defined from it.

Conditionally on the other channels, every weighted channel is used by all ten models of every profile, under both weightings, but unevenly. In HUST, permuting mean discharge voltage raises the error more than permuting capacity does. Conditional permutation of charge time raises it by only 0.015 [0.012, 0.018], against 0.068 for capacity and 0.630 for mean discharge voltage, so part of the ground truth lies on a channel the model reads only weakly. Permuting a null channel raises the error by at most 0.0003 at the upper end of its interval, in every profile. Trained models therefore do not use the null channels, and any attribution they receive is error of the method.

Only FORM carries inserted patterns, and a trained model is not shown to learn them. At their fitted rate and amplitude, the pattern term carries 1.7×10^{-5} of the variance of the mean term. Removing the mean term from the training target raises the model's error on y by 2.58 [2.23, 3.03], whereas removing the pattern term leaves it unchanged under both weightings. Scores of the sparse set on trained models of FORM would therefore describe a term the model has not learned, and are not reported. The sparse set is scored only where the map is known to contain it, on the reference model and on degraded maps of the ground truth.

4.3 Responsiveness of the Scores

In each profile, the ground-truth field is degraded on 20 windows drawn at random from each of the 45 test units, 900 windows in all, under recency and uniform weighting. Every score takes its ideal value on the undegraded field, 1 in rank agreement and 0 in both errors. On a fully permuted map, under recency weighting, it is about 0 in rank agreement, 0.46 to 0.63 in channel allocation error and 5.05 to 5.43 positions in temporal profile error.

Rank agreement responds to added noise and to permuted cells, but barely to smoothing or to a partial shift of mass toward either end of the window (Table 4, Figure 7). Smoothing with a standard deviation of eight positions changes it

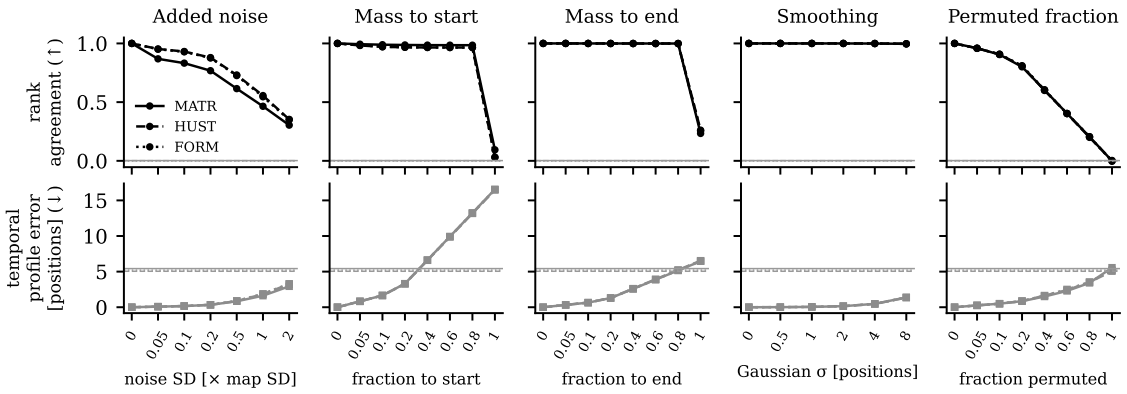


Figure 7: Response of rank agreement (top) and of the temporal profile error (bottom) to degradation of the ground truth, under recency weighting, averaged over test units; MATR solid, HUST dashed, FORM dotted. Thin grey lines: score of a fully permuted map.

Table 4: Change of each score under degradation of the ground truth, under recency weighting, as a fraction of the distance between its values on the undegraded and on a fully permuted map (0: unchanged; 1: as far as a random permutation). In parentheses, the magnitude: added noise in standard deviations of the map, moved mass and permuted cells as a fraction, smoothing as the Gaussian standard deviation in positions. Rank: rank agreement; alloc.: channel allocation error; temporal: temporal profile error.

	MATR			HUST			FORM		
Degradation	rank	alloc.	temporal	rank	alloc.	temporal	rank	alloc.	temporal
Added noise (2)	0.70	0.88	0.54	0.65	0.86	0.64	0.65	0.85	0.63
Mass to start (0.8)	0.02	0.00	2.44	0.04	0.00	2.61	0.03	0.00	2.54
Mass to end (0.8)	0.00	0.00	0.95	0.00	0.00	1.04	0.00	0.00	0.99
Smoothing (8)	0.00	0.00	0.25	0.00	0.00	0.27	0.00	0.01	0.26
Permuted cells (0.8)	0.80	0.80	0.63	0.80	0.82	0.70	0.79	0.80	0.68

by at most 0.004 of its distance to chance, and moving 80% of each channel's mass toward the end by at most 0.001. It falls only when all of the mass is moved. The temporal profile error responds to every degradation. The same smoothing moves it about a quarter of the way to chance and the same shift about the whole way. A shift toward the start takes it more than twice as far from the ground truth as a random permutation does, since under recency weighting the graded field is concentrated at the end of the window. The channel allocation error responds only to the two degradations that move mass between channels, added noise and permuted cells. Under uniform weighting the scores respond in the same way, except to smoothing. Smoothing leaves the nearly constant uniformly weighted field almost unchanged and moves no score by more than 0.03 of its distance to chance.

In FORM, retrieval of the sparse set is measured on the windows that contain at least one of its cells. Scored on the window paired with its pattern-free counterpart, the undegraded field retrieves the sparse set perfectly, with an average precision of 1 against a chance level of 0.07. Scored on the window alone, it reaches only 0.095. Added noise and permuted cells lower the paired score to 0.37 and 0.29 already at their smallest magnitudes, since both are scaled to the whole map, against which the patterns are small. Moving 80% of the mass lowers it to 0.72–0.73 and moving all of it to about 0.1. Smoothing barely changes it, since the difference between two smoothed maps still peaks where the patterns are. Under uniform weighting the paired score responds in the same way.

4.4 Reference Values for Published Methods

Table 5 and Figure 8 give the scores of the three methods under recency weighting. On the reference model, all three return the exact attribution, with an absolute difference of at most 2.5×10^{-5} against largest map values of 0.02 to 0.08. They therefore reach the ceiling, 0.849 to 0.957 in rank agreement. In FORM, the difference between each method's maps on a window and on its counterpart also retrieves the sparse set exactly, with an average precision of 1.

On the trained model, every method lies below the ceiling in every score. In every score of every profile but one, the largest difference between methods is smaller than the range of at least one method over the ten models. The

Table 5: Reference values of each attribution method on the primary trained model and on the reference model, under recency weighting, on 120 windows from the 45 test units of each profile. Unit means with 95% bootstrap intervals, given on the reference model for the exact attribution only. TimeSHAP rows use coalition-sampling seed 0. Range: smallest and largest unit mean over the ten equally accurate models. TimeSHAP, average event: run from the average event and scored against the ground truth for that baseline.

	Rank agreement		Allocation error		Temporal error		Zero-weight mass	
Method	trained	reference	trained	reference	trained	reference	trained	reference
MATR								
IG	0.485 [0.445, 0.519]	0.869	0.294 [0.277, 0.314]	0.075	1.86 [1.68, 2.03]	0.38	0.134 [0.117, 0.153]	0.000
Occlusion	0.420 [0.368, 0.466]	0.869	0.377 [0.353, 0.407]	0.075	2.56 [2.39, 2.76]	0.38	0.198 [0.171, 0.233]	0.000
TimeSHAP	0.493 [0.454, 0.527]	0.870	0.301 [0.283, 0.321]	0.075	1.82 [1.65, 2.00]	0.38	0.136 [0.120, 0.155]	0.000
Exact attribution	–	0.869 [0.817, 0.909]	–	0.075 [0.060, 0.097]	–	0.38 [0.31, 0.45]	–	0.000
Range, IG	0.406–0.506	–	0.287–0.341	–	1.86–2.73	–	–	–
Range, Occlusion	0.326–0.449	–	0.376–0.414	–	2.32–3.36	–	–	–
Range, TimeSHAP	0.415–0.514	–	0.292–0.343	–	1.82–2.55	–	–	–
TimeSHAP, average event	0.527 [0.488, 0.558]	–	0.313 [0.292, 0.341]	–	2.21 [2.07, 2.35]	–	0.138 [0.122, 0.166]	–
HUST								
IG	0.734 [0.700, 0.758]	0.957	0.164 [0.151, 0.179]	0.063	2.82 [2.65, 3.00]	0.18	0.037 [0.031, 0.051]	0.000
Occlusion	0.750 [0.715, 0.776]	0.957	0.182 [0.166, 0.203]	0.063	2.93 [2.77, 3.10]	0.18	0.040 [0.032, 0.057]	0.000
TimeSHAP	0.739 [0.705, 0.764]	0.957	0.176 [0.161, 0.191]	0.063	2.86 [2.71, 3.03]	0.18	0.038 [0.032, 0.052]	0.000
Exact attribution	–	0.957 [0.922, 0.978]	–	0.063 [0.051, 0.079]	–	0.18 [0.14, 0.24]	–	0.000
Range, IG	0.597–0.747	–	0.125–0.218	–	2.63–3.80	–	–	–
Range, Occlusion	0.566–0.772	–	0.158–0.257	–	2.81–3.54	–	–	–
Range, TimeSHAP	0.600–0.781	–	0.136–0.230	–	2.61–3.81	–	–	–
TimeSHAP, average event	0.697 [0.664, 0.721]	–	0.156 [0.142, 0.174]	–	2.85 [2.71, 2.99]	–	0.030 [0.026, 0.041]	–
FORM								
IG	0.770 [0.723, 0.806]	0.849	0.215 [0.197, 0.240]	0.125	3.89 [3.72, 4.12]	0.63	0.039 [0.033, 0.045]	0.000
Occlusion	0.744 [0.696, 0.783]	0.849	0.265 [0.244, 0.289]	0.125	3.71 [3.49, 4.05]	0.63	0.048 [0.042, 0.055]	0.000
TimeSHAP	0.771 [0.725, 0.808]	0.850	0.220 [0.203, 0.245]	0.125	3.82 [3.65, 4.04]	0.63	0.039 [0.033, 0.045]	0.000
Exact attribution	–	0.849 [0.795, 0.891]	–	0.125 [0.103, 0.165]	–	0.63 [0.53, 0.79]	–	0.000
Range, IG	0.438–0.776	–	0.196–0.257	–	2.97–3.89	–	–	–
Range, Occlusion	0.431–0.753	–	0.229–0.271	–	3.19–3.71	–	–	–
Range, TimeSHAP	0.458–0.779	–	0.194–0.262	–	2.82–3.82	–	–	–
TimeSHAP, average event	0.740 [0.696, 0.784]	–	0.247 [0.222, 0.273]	–	3.57 [3.35, 3.84]	–	0.047 [0.041, 0.054]	–

choice of model thus moves the scores as much as the choice of method does (Figure 8). In rank agreement, for instance, the methods differ by at most 0.073, whereas each method's range is at least 0.099, and 0.32 to 0.34 in FORM. The exception is channel allocation in MATR. There, feature occlusion is further from the ground truth than IG and TimeSHAP, 0.377 against 0.294 and 0.301, and each of its ten models lies above every model of the other two methods. Under uniform weighting, where the ten models are run with IG and occlusion only, the difference of 0.072 still exceeds their ranges, of at most 0.071. The share of attribution on zero-weight channels, where $\phi^\star$ is zero, is highest in MATR, 0.134 to 0.198, where two of them, internal resistance and temperature, carry the degradation state. It is at most 0.048 in HUST and FORM, whose zero-weight channels are the null channels.

Two choices move the values: the baseline and the weighting. Run from the average event, TimeSHAP's values move by up to 0.04 in rank agreement and 0.4 positions in temporal profile error, in different directions in different profiles. In HUST and FORM this is enough to change its place among the methods in rank agreement, so a reference value holds only for the baseline from which it was computed. Under uniform weighting, on the models trained for that weighting, rank agreement changes by at most 0.04 in MATR and HUST but falls in FORM, to 0.532–0.570 against a ceiling of 0.826. The allocation error changes by at most 0.013. The temporal profile error rises to 3.15–5.47 positions, more than the 1.70–2.59 positions that a fully permuted ground truth scores under that weighting. When every position is equally relevant, every method on a trained model thus distributes attribution along the window further from the ground truth than a random permutation of the ground truth does.

5 Discussion

The profiles reproduce the fitted properties of their datasets up to their spreads. From the HUST profile, a model learns the dependence between the sequences and the RUL as well as from the measured cells. From the MATR and FORM profiles, it learns part of that dependence. The remaining part lies in the spreads or in structure that the compared properties do not summarize. A generator written down as explicit components is not expected to reproduce every property of a measured process. This is the trade-off between realism and ground-truth availability that existing

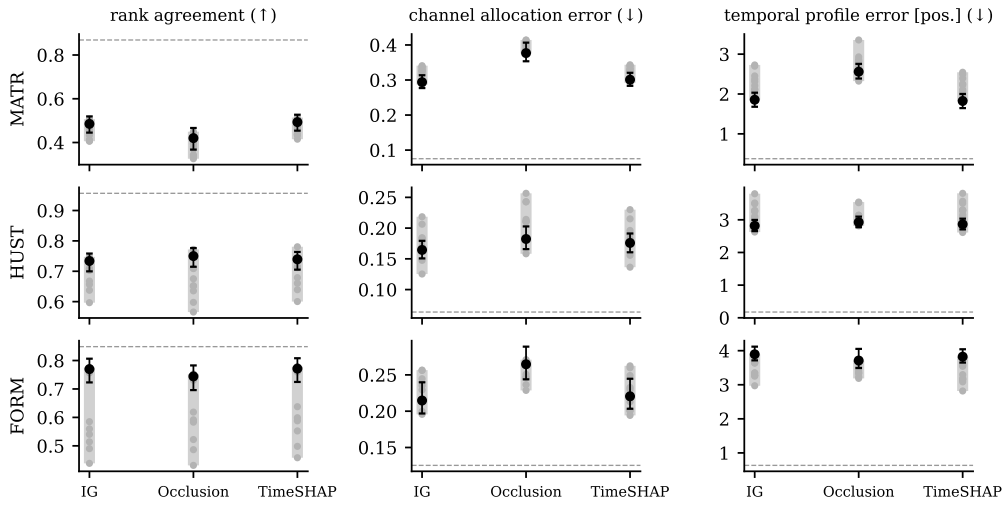


Figure 8: Scores of each method on the trained models, under recency weighting: unit mean on the primary model with its 95% interval (black), unit means of the ten equally accurate models (grey points) and their range (grey band), and the exact attribution on the reference model (dashed).

benchmarks acknowledge without measuring it [16], and the profiles measure how far it extends. One candidate for the missing structure follows from the construction. A mapping shared by all units cannot express channel trajectories whose shape differs between groups of cells. The cells of MATR and FORM were charged or formed under protocols that differ from cell to cell, whereas those of HUST share one charging protocol. Whatever the cause, a method evaluated on MATR or FORM has not been shown to behave the same way on a model that also reads the structure the generator omits.

The ground truth is usable, but reading a channel does not fix how a model divides the target among channels. Every model reads every weighted channel, plausibly because the per-unit offsets, which enter y, are not fully predicted by the other channels. No part of $\phi^\star$ therefore lies on a channel the model ignores. How much a model relies on a channel, however, does not follow the share of the target assigned to it. In HUST, charge time has the same share as the other two weighted channels but is read only weakly. This freedom reappears in the attribution scores of trained models.

The capacity recoveries of FORM occur often enough to be told apart from noise, and on the reference model every method retrieves them exactly. They carry a negligible part of the target, however, so a trained model has no reason to learn them, and localization cannot be tested on it. Two observations bear on how such effects are scored beside a graded field. On a single window the graded field masks the patterns, so retrieval stays near chance even for the undegraded ground truth. It becomes informative only on the difference between a window and its pattern-free counterpart, so retrieval metrics established on planted classification data cannot be applied unchanged where a graded field dominates. Even this paired score falls sharply under errors scaled to the whole map, so a map close to the graded field can still fail to localize the patterns. Testing localization on a trained model needs data whose localized effects carry enough of the target for a model to use them. Raising the amplitudes of FORM would supply such data, but the profile would then no longer stand for its dataset.

For a graded ground truth, agreement in order is not agreement in shape. Smoothing a map or moving part of its mass toward the end preserves the order on which rank agreement depends, although both are plausible failures of an attribution method. The temporal profile error and the channel allocation error measure what rank agreement leaves out, where along the window and on which channel attribution is placed. A graded ground truth therefore requires them beside rank agreement.

On the reference model, the three methods reach the ceiling, as any method exact for an affine model must. The reference model is therefore a correctness check rather than a ranking: a method that fell below the ceiling would be shown to misattribute a known function. It is also the only model on which a score measures the method alone, since only there is the function that defines the decomposable target the model itself. On trained models, equally accurate models differ more than the methods do. This is expected when the channels share one degradation state, since a faithful attribution of the function a model learned can then differ from $\phi^\star$. The trained-model values therefore describe what a method returns on one model that fits the target, which is what a practitioner will have, rather than which method is more faithful.

Positions appear to behave like channels in this respect. Under uniform weighting, every method on a trained model places attribution along the window further from $\phi^\star$ than a random permutation does. This is consistent with neighboring positions carrying nearly the same degradation state, so that a model need not spread its dependence over the window as the target does. The one difference between methods that exceeds the range of the ten models, channel allocation in MATR, arises in the only profile whose zero-weight channels carry the degradation state. On the primary model, occlusion also places the largest share of attribution on those channels. Part of the difference may thus lie in how the methods treat channels that carry the degradation state but have no weight in the target. Since a change of baseline alone can reorder the methods, reference values are comparable only between methods run from the same baseline.

5.1 Limitations and Future Work

Several design choices bound these conclusions. The weighted channels have equal shares of the target, so nothing here shows how the scores behave when one channel dominates. The EOL fraction ρ and the window length L are fixed. The models are of one architecture, with ten models that vary initialization and size only, so their range may understate the variation among equally accurate models. The within-unit correlation that sets the noise correlation is estimated from residuals about a degradation state read from noisy capacity. Capacity noise therefore inflates the correlations of capacity with the other channels. The generator reproduces this bias and re-estimation adds it again, so in MATR and FORM the capacity pairs depart from the fitting split more in generated units than in held-out cells (Table 1). In every profile, generated units are less noisy than the cells they stand for, so the scores are obtained at a lower noise level than the measured data carry. In FORM, 21 of the 300 generated units on which the properties are estimated end above the EOL threshold by more than the tolerance of $0.01\, q_{\mathrm{nom}}$ and are left out of the property rows that depend on EOL.

Whether the transfer ratios of MATR and FORM come from channel trajectories that differ in shape between protocols can be tested by fitting a mapping per protocol group. This comes at the cost of levels set by the protocol, which enter the decomposable target through the weights. The generator also fixes three components and a parametric family for each, which bounds the shapes a profile can express. Learning the components instead, with a model emitting a smooth, a localized and an unstructured part that sum to the sequence, would improve the realism of x while keeping the attribution ground truth exact. The ground truth requires only that the generator record which part of each value it produced, not that the split be recovered from measurement. Such a model would have to retain the three roles, against which the graded field, the sparse set and the exclusion of noise are defined. The present testbed, in which a known decomposition exists, would serve to validate it. Finally, ranking methods on trained models requires an attribution ground truth that is identifiable for them, at the level of both channels and positions, which the shared degradation state prevents in the present construction.

6 Conclusion

DEGRADX supplies attribution ground truth for a continuous target under sustained mean-nonstationarity, defined at every observation rather than at planted positions. Each profile comes with a measurement of how closely it reproduces the dataset it was fitted to. The profiles reproduce the fitted properties of their datasets up to their spreads. In transfer, a model trained on the HUST profile is not distinguished from one trained on the measured cells. Trained on the MATR or FORM profile, a model reaches about one and a half times that error, so these profiles reproduce part of the dependence between the sequences and the remaining useful life. The ground truth is usable, since the target is learnable and every channel that enters it is read by trained models. The one localized effect in the data, a capacity recovery in FORM, carries too little of the target for a trained model to be shown to learn it. Rank agreement misses smoothing and partial shifts of an attribution map, which the temporal profile error registers, while the channel allocation error registers attribution placed on the wrong channel. On the reference model, the three methods recover the exact attribution. On trained models, equally accurate models differ more than the methods do, so a trained-model score says what a method returns on one model that fits the target, not which method is more faithful. In practice, the testbed can be used to check whether an attribution method recovers a known attribution before the method is applied to a model of a degrading system. It also shows which failures of an attribution map a score registers, and whether a difference between methods exceeds the variation among equally accurate models. Learning the channel trajectories per protocol, finding degradation data whose localized effects carry part of the target, and making the ground truth identifiable on trained models are the steps that remain.

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References

[1] Douglas Thomas and Brian Weiss. Maintenance costs and advanced maintenance techniques: Survey and analysis. *International Journal of Prognostics and Health Management*, 12(1), 2021. doi: 10.36001/ijphm.2021.v12i1.2883.

[2] Sajad Moosavi, Maryam Farajzadeh-Zanjani, Roozbeh Razavi-Far, Vasile Palade, and Mehrdad Saif. Explainable AI in manufacturing and industrial cyber–physical systems: A survey. *Electronics*, 13(17), 2024. doi: 10.3390/electronics13173497.

[3] Hugues Turbé, Mina Bjelogrlic, Christian Lovis, and Gianmarco Mengaldo. Evaluation of post-hoc interpretability methods in time-series classification. *Nature Machine Intelligence*, 5(3):250–260, 2023. doi: 10.1038/s42256-023-00620-w.

[4] Davide Italo Serramazza, Thach Le Nguyen, and Georgiana Ifrim. Improving the evaluation and actionability of explanation methods for multivariate time series classification. In *Machine Learning and Knowledge Discovery in Databases. Research Track*, pages 177–195. Springer Nature Switzerland, 2024. doi: 10.1007/978-3-031-70359-1_11.

[5] Ilija Šimić, Eduardo Veas, and Vedran Sabol. A comprehensive analysis of perturbation methods in explainable AI feature attribution validation for neural time series classifiers. *Scientific Reports*, 15(1):26607, 2025. doi: 10.1038/s41598-025-09538-2.

[6] Gregor Baer, Isel Grau, Chao Zhang, and Pieter Van Gorp. Why do class-dependent evaluation effects occur with time series feature attributions? A synthetic data investigation. In *Machine Learning and Principles and Practice of Knowledge Discovery in Databases*, pages 380–395. Springer Nature Switzerland, 2026. doi: 10.1007/978-3-032-19105-2_27.

[7] Aya Abdelsalam Ismail, Mohamed Gunady, Héctor Corrada Bravo, and Soheil Feizi. Benchmarking deep learning interpretability in time series predictions. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, pages 6441–6452, 2020.

[8] Sana Tonekaboni, Shalmali Joshi, Kieran R Campbell, David Duvenaud, and Anna Goldenberg. What went wrong and when? Instance-wise feature importance for time-series black-box models. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, pages 799–809, 2020.

[9] Owen Queen, Thomas Hartvigsen, Teddy Koker, Huan He, Theodoros Tsiligkaridis, and Marinka Zitnik. Encoding time-series explanations through self-supervised model behavior consistency. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS '23, pages 32129–32159, 2023.

[10] Gregor Baer. xaitimesynth: A Python package for evaluating attribution methods for time series with synthetic ground truth. *arXiv preprint arXiv:2603.06781*, 2026.

[11] Qilin Qu, Lei Zhang, and Ying Zheng. Explainable AI for remaining useful life prediction in industrial systems: a survey. *Measurement Science and Technology*, 37(4), 2026. doi: 10.1088/1361-6501/ae319f.

[12] Jonathan Crabbé and Mihaela van der Schaar. Explaining time series predictions with dynamic masks. *arXiv preprint arXiv:2106.05303*, 2021.

[13] Kristen A. Severson, Peter M. Attia, Norman Jin, Nicholas Perkins, Benben Jiang, Zi Yang, Michael H. Chen, Muratahan Aykol, Patrick K. Herring, Dimitrios Fraggedakis, Martin Z. Bazant, Stephen J. Harris, William C. Chueh, and Richard D. Braatz. Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy*, 4(5):383–391, 2019. doi: 10.1038/s41560-019-0356-8.

[14] Guijun Ma, Songpei Xu, Benben Jiang, Cheng Cheng, Xin Yang, Yue Shen, Tao Yang, Yunhui Huang, Han Ding, and Ye Yuan. Real-time personalized health status prediction of lithium-ion batteries using deep transfer learning. *Energy & Environmental Science*, 15(10):4083–4094, 2022. doi: 10.1039/d2ee01676a.

[15] Xiao Cui, Stephen Dongmin Kang, Sunny Wang, Justin A. Rose, Huada Lian, Alexis Geslin, Steven B. Torrisi, Martin Z. Bazant, Shijing Sun, and William C. Chueh. Data-driven analysis of battery formation reveals the role of electrode utilization in extending cycle life. *Joule*, 8(11):3072–3087, 2024. doi: 10.1016/j.joule.2024.07.024.

[16] Yang Liu, Sujay Khandagale, Colin White, and Willie Neiswanger. Synthetic benchmarks for scientific research in explainable machine learning. In *Advances in Neural Information Processing Systems Datasets Track*, 2021.

[17] Joseph Enguehard. Learning perturbations to explain time series predictions. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23, pages 9329–9342, 2023.

[18] Zichuan Liu, Yingying Zhang, Tianchun Wang, Zefan Wang, Dongsheng Luo, Mengnan Du, Min Wu, Yi Wang, Chunlin Chen, Lunting Fan, and Qingsong Wen. Explaining time series via contrastive and locally sparse perturbations. In *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[19] Jacqueline Höllig, Steffen Thoma, and Florian Grimm. XTSC-Bench: Quantitative benchmarking for explainers on time series classification. In *2023 International Conference on Machine Learning and Applications (ICMLA)*, pages 1126–1131, 2023. doi: 10.1109/ICMLA58977.2023.00168.

[20] Maciej Mozolewski, Szymon Bobek, and Grzegorz J. Nalepa. ExplainTS: a benchmark dataset of pretrained models and post-hoc explanations for time-series classification. *Frontiers in Artificial Intelligence*, 9, 2026. doi: 10.3389/frai.2026.1759110.

[21] Domjan Barić, Petar Fumić, Davor Horvatić, and Tomislav Lipic. Benchmarking attention-based interpretability of deep learning in multivariate time series predictions. *Entropy*, 23(2), 2021. doi: 10.3390/e23020143.

[22] Muhammad Hasan Ferdous, Emam Hossain, and Md Osman Gani. TimeGraph: Synthetic benchmark datasets for robust time-series causal discovery. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, KDD '25, page 5425–5435. Association for Computing Machinery, 2025. doi: 10.1145/3711896.3737439.

[23] Harshavardhan Kamarthi, Shangqing Xu, Xinjie Tong, Xingyu Zhou, James Peters, Joseph Czyzyk, and B. Aditya Prakash. Hierarchical industrial demand forecasting with temporal and uncertainty explanations. *arXiv preprint arXiv:2603.06555*, 2026.

[24] Matthias Hertel, Sebastian Pütz, Ralf Mikut, Veit Hagenmeyer, and Benjamin Schäfer. Explainable time-series forecasting with sampling-free SHAP for transformers. *Nature Communications*, 17:4723, 2026. doi: 10.1038/s41467-026-73243-5.

[25] Peiyu Yang, Naveed Akhtar, Jiantong Jiang, and Ajmal Mian. A backdoor-based explainable AI benchmark for high fidelity evaluation of attributions. *arXiv preprint arXiv:2405.02344*, 2025.

[26] Hamidreza Eivazi, André Hebenbrock, Raphael Ginster, Steffen Blömeke, Stefan Wittek, Christoph Hermann, Thomas S. Spengler, Thomas Turek, and Andreas Rausch. DiffBatt: A diffusion model for battery degradation prediction and synthesis. In *NeurIPS 2024 Workshop Foundation Models for Science: Progress, Opportunities, and Challenges*, 2024.

[27] Falak Naaz, Aniruddh Herle, Janamejaya Channegowda, Aditya Raj, and Meenakshi Lakshminarayanan. A generative adversarial network-based synthetic data augmentation technique for battery condition evaluation. *International Journal of Energy Research*, 45(13):19120–19135, 2021. doi: 10.1002/er.7013.

[28] Meghana Sudarshan, Jaya Vikeswara Rao Vajja, and Vikas Tomar. DegradAI: A scalable framework for early battery health diagnosis from limited data. *npj Clean Energy*, 1(8), 2025. doi: 10.1038/s44406-025-00008-2.

[29] Cristóbal Esteban, Stephanie L. Hyland, and Gunnar Rätsch. Real-valued (medical) time series generation with recurrent conditional GANs. *arXiv preprint arXiv:1706.02633*, 2017.

[30] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.

[31] Paul Jeha, Michael Bohlke-Schneider, Pedro Mercado, Shubham Kapoor, Rajbir Singh Nirwan, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. PSA-GAN: Progressive self attention GANs for synthetic time series. In *Proceedings of the 10th International Conference on Learning Representations*, 2022.

[32] Myroslav Mishchuk, Rafał Cupek, and Olena Pavliuk. Explaining LSTM battery RUL prediction via temporal attribution. In *Intelligent Information and Database Systems*, pages 216–231. Springer Nature Singapore, 2026. doi: 10.1007/978-981-92-0071-9_15.

[33] Wei He, Nicholas Williard, Michael Osterman, and Michael Pecht. Prognostics of lithium-ion batteries based on Dempster–Shafer theory and the Bayesian Monte carlo method. *Journal of Power Sources*, 196(23):10314–10321, 2011. doi: 10.1016/j.jpowsour.2011.08.040.

[34] Saurabh Saxena, Logan Ward, Joseph Kubal, Wenquan Lu, Susan Babinec, and Noah Paulson. A convolutional neural network model for battery capacity fade curve prediction using early life data. *Journal of Power Sources*, 542:231736, 2022. doi: 10.1016/j.jpowsour.2022.231736.

[35] Haipeng Pan, Chengte Chen, and Minming Gu. A method for predicting the remaining useful life of lithium batteries considering capacity regeneration and random fluctuations. *Energies*, 15(7), 2022. doi: 10.3390/en15072498.

[36] Taichun Qin, Shengkui Zeng, Jianbin Guo, and Zakwan Skaf. A rest time-based prognostic framework for state of health estimation of lithium-ion batteries with regeneration phenomena. *Energies*, 9(11), 2016. doi: 10.3390/en9110896.

[37] Benjamín E. Olivares, Matías A. Cerda Munoz, Marcos E. Orchard, and Jorge F. Silva. Particle-filtering-based prognosis framework for energy storage devices with a statistical characterization of state-of-health regeneration phenomena. *IEEE Transactions on Instrumentation and Measurement*, 62(2):364–376, 2013. doi: 10.1109/TIM.2012.2215142.

[38] Qiuhui Ma, Ying Zheng, Weidong Yang, Yong Zhang, and Hong Zhang. Remaining useful life prediction of lithium battery based on capacity regeneration point detection. *Energy*, 234:121233, 2021. doi: 10.1016/j.energy.2021.121233.

[39] João Bento, Pedro Saleiro, André F. Cruz, Mário A.T. Figueiredo, and Pedro Bizarro. TimeSHAP: Explaining recurrent models through sequence perturbations. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, KDD '21, page 2565–2573. Association for Computing Machinery, 2021. doi: 10.1145/3447548.3467166.

[40] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML'17, page 3319–3328. JMLR.org, 2017.

[41] Harini Suresh, Nathan Hunt, Alistair Johnson, Leo Anthony Celi, Peter Szolovits, and Marzyeh Ghassemi. Clinical intervention prediction and understanding using deep networks. *arXiv preprint arXiv:1705.08498*, 2017.